

Zoning and the Dynamics of Urban Redevelopment

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Zoning is often denounced as a culprit for high housing prices

- Many cities are upzoning (Cambridge, NYC, Austin, Seattle, Portland, Sacramento, Minneapolis, Raleigh...)

THE WALL STREET JOURNAL.

**Only Zoning Reform Can Solve
America's Housing Crisis**

Los Angeles Times

**L.A. can't become an affordable city
by protecting single-family zoning**

FT

**FINANCIAL
TIMES**

**Planning rules are driving the
global housing crisis**

The Boston Globe

**Do homes really need 2-acre lots?
Spoiler alert: No.**

Technological constraints matter too

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- Change comes from **redevelopment**.
 - Demolishing and replacing old buildings.
- Increasingly how cities grow (Frolking et al., 2024).
- Requires paying large **fixed costs**.



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- **Quasi-experimental evidence** from recent zoning changes.

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Policy application: What would be the effects of relaxing zoning?

- **Quasi-experimental evidence** from recent zoning changes.
- Use model to predict the **effects of relaxing zoning** on **construction** and **prices**.

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Findings

- Zoning strongly hinders construction, and relaxing zoning yields **large welfare gains**.
- But **adjustment** to policy change is **slow**, **take-up of upzoning is limited**.
- Relaxing zoning most effective where **prices are high/density is low**.
- Affordability **gains geographically spread out** due to migration.

Data and stylized facts

Data

- **Quantities:** Panel of **land uses at the parcel level** covering 2004–2022.
 - Assembled using cadastral maps, tax records, StreetEasy, building permits, certificates of occupancy, etc.
 - 833,000 parcels, 22,000 redevelopment events.

Data

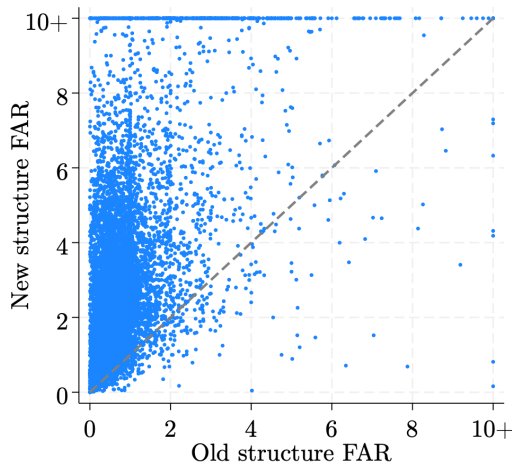
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- **Zoning:** FAR limits for each parcel and year.

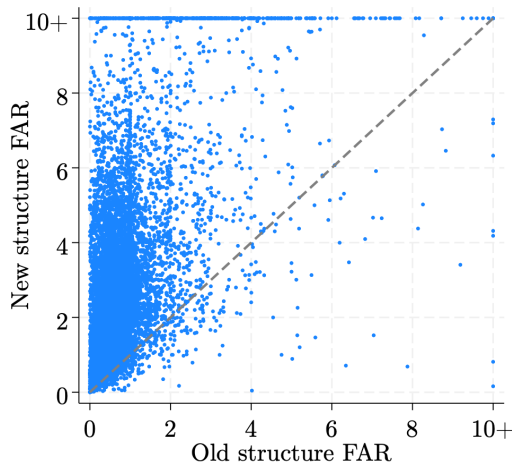
Redevelopment means densification

- New buildings 3.4 times larger than the ones they replace (on average).



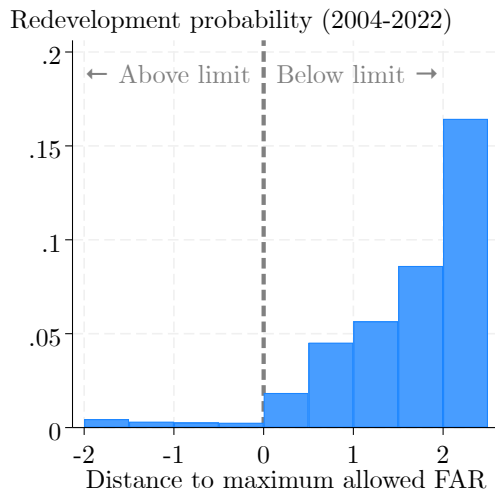
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- Suggests large redevelopment **fixed costs**.



Redevelopment mostly happens when upward growth is allowed

- Buildings at/over the zoning limit are seldom redeveloped.



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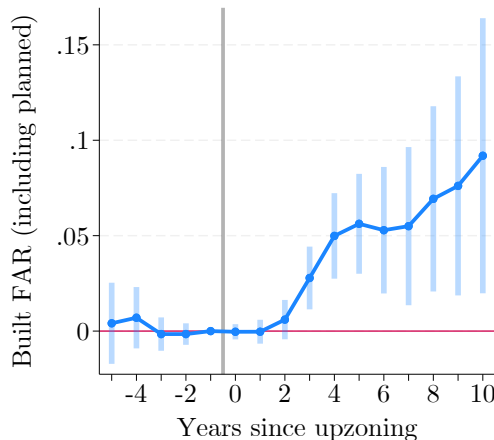
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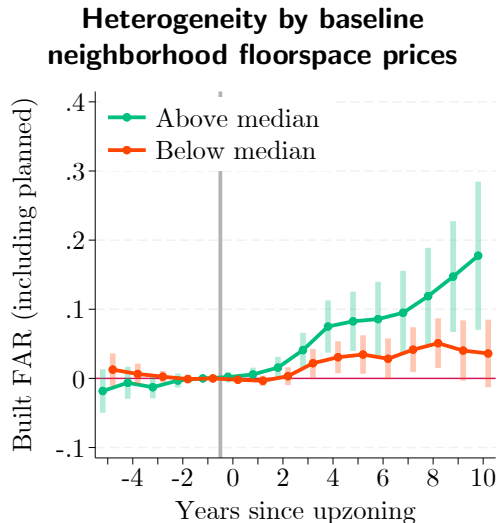
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- $\approx 10\%$ of increase used after a decade.



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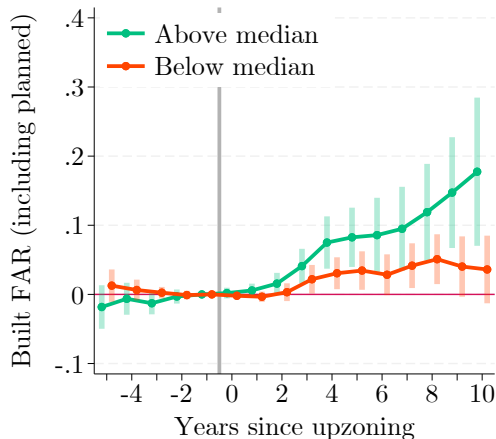
Effects concentrate in high-price neighborhoods and underbuilt parcels



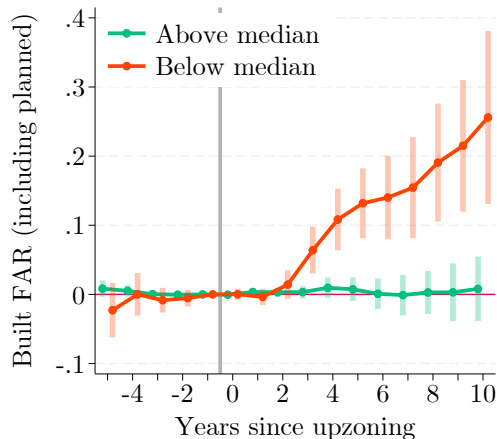
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Heterogeneity by baseline neighborhood floorspace prices



Heterogeneity by baseline built FAR (within neighborhood)



Remaining questions

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 - ② What are the effects of allowing more construction on **prices**?
- Answer these questions with a **model of urban change**.

A model of urban change

Supply of floorspace

Behavior of developers given prices and regulation

Developer choices

Developers are atomistic and perfectly foresighted. They make decisions each year:

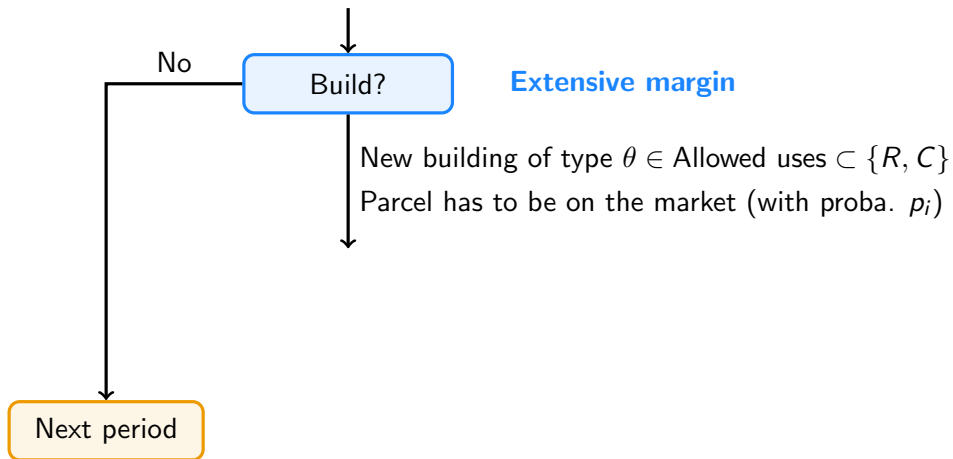
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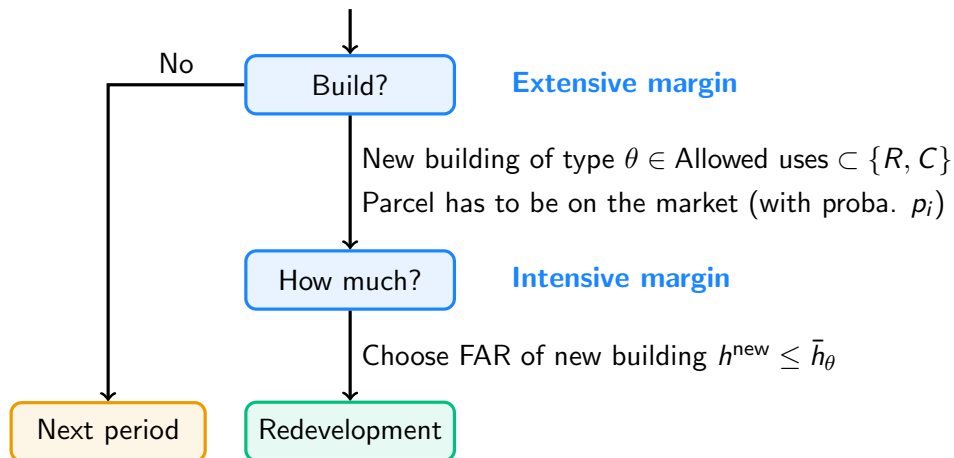
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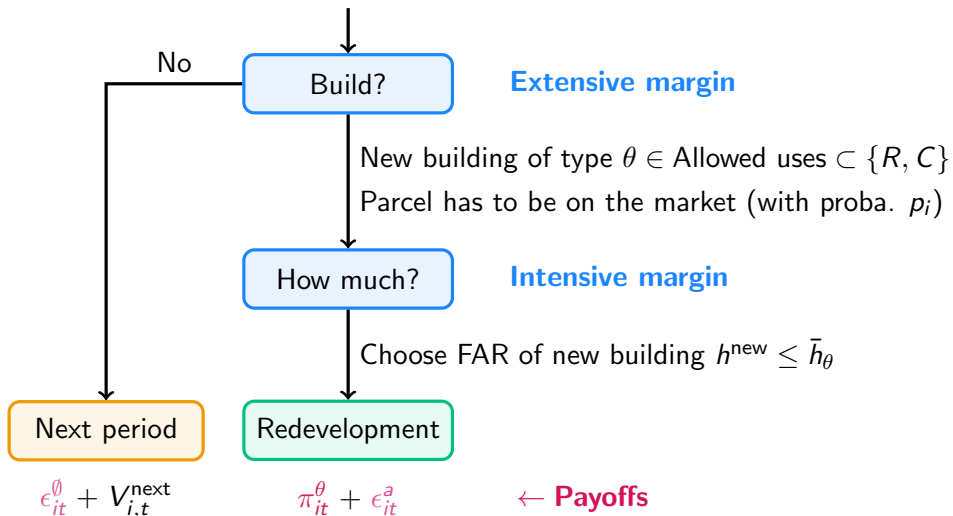
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Developer profit function

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- **Variable costs (VC)**: construction costs, increase with size of new building.
- **Fixed costs (FC)**: eviction, demolition, permitting, etc.
 - Vary by building (e.g., larger for bigger buildings in dense neighborhoods).

Demand for floorspace

Evolution of prices given developers' choices

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- Features agglomeration, migration, congestion.

Model estimation and validation

Supply-side parameters

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Prices and variable costs

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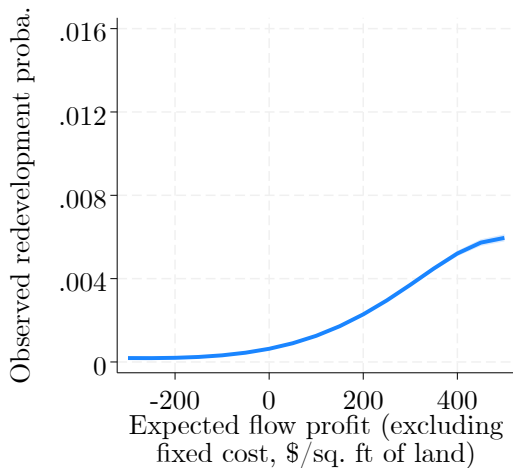
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- **Variable costs:** Revealed preferences.
 - Use data on FARs chosen by developers for new buildings + prices/zoning they faced.
 - Recover costs under the assumption that developers choose FARs to maximize profit under the zoning constraint.

Fixed costs estimation

- **Idea:**

- Compute expected profit from redevelopment, excluding fixed costs.
- Find fixed costs to best match relationship with redevelopment probability.



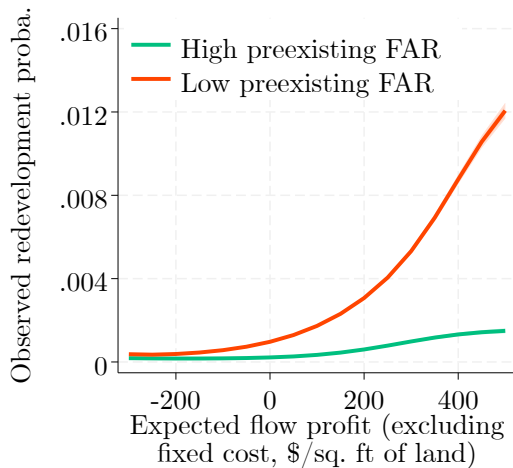
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- **Parameterization:** fixed costs

- Increase with size of old building.
- Increase with neighborhood density.
- Larger in historic districts.



Model validation

Predicted effect of upzonings

- **Validation exercise:**

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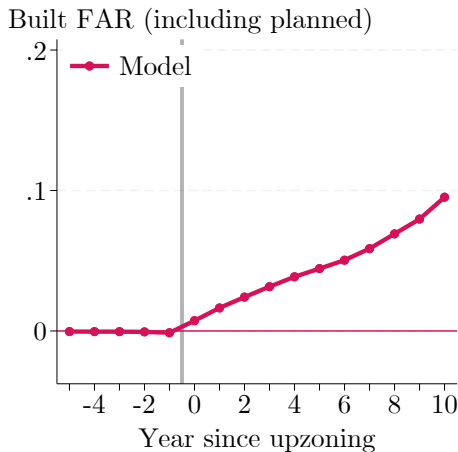
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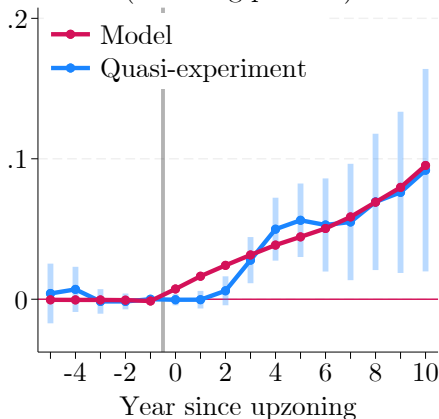


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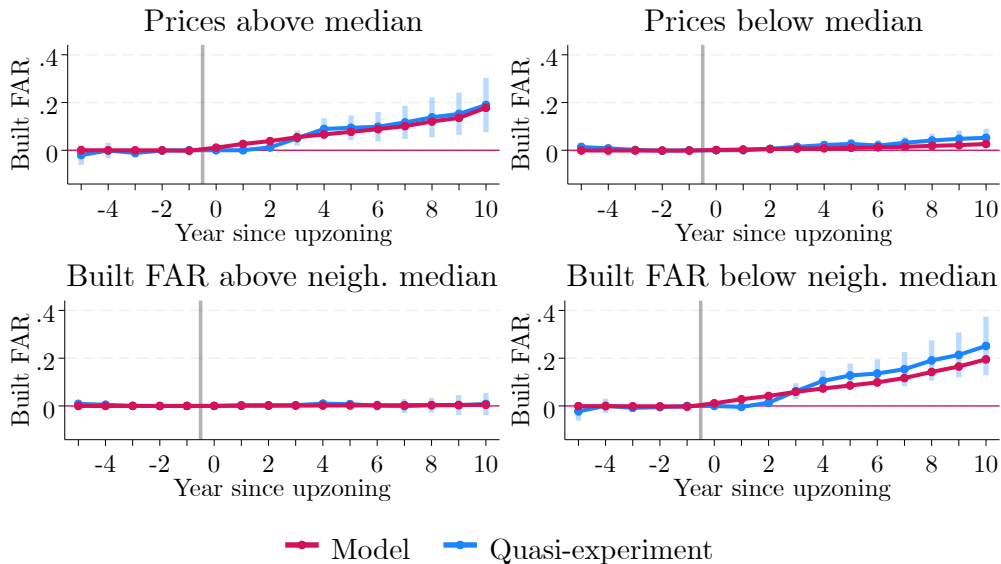
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- 4 Compare with quasi-experimental estimates.

Built FAR (including planned)



Model validation: Predicted effect of upzonings



Implications for zoning policy

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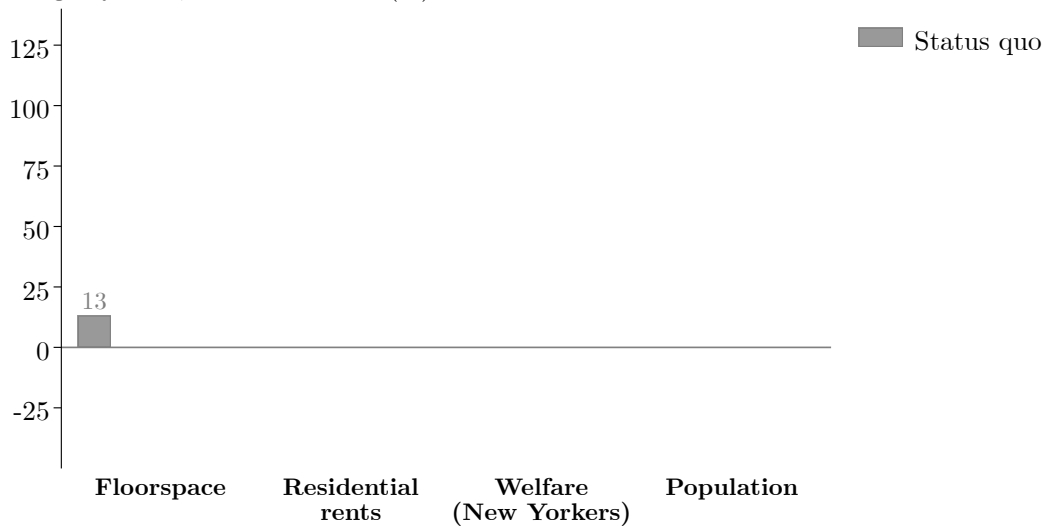
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 - ④ **Frictionless benchmark** (no zoning, no adjustment costs, price = marginal cost).
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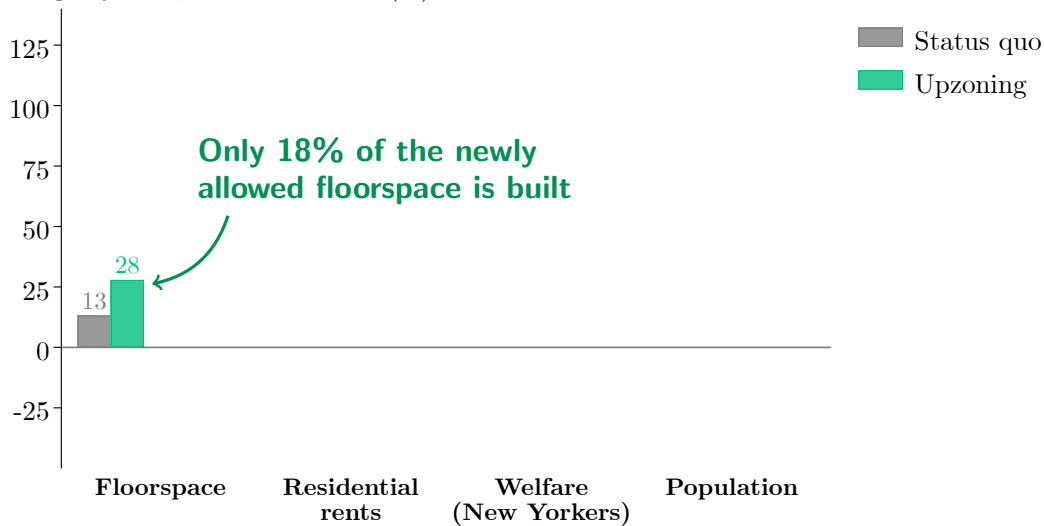
Under current zoning, NYC continues to grow slowly

Change by 2060, relative to 2019 (%)



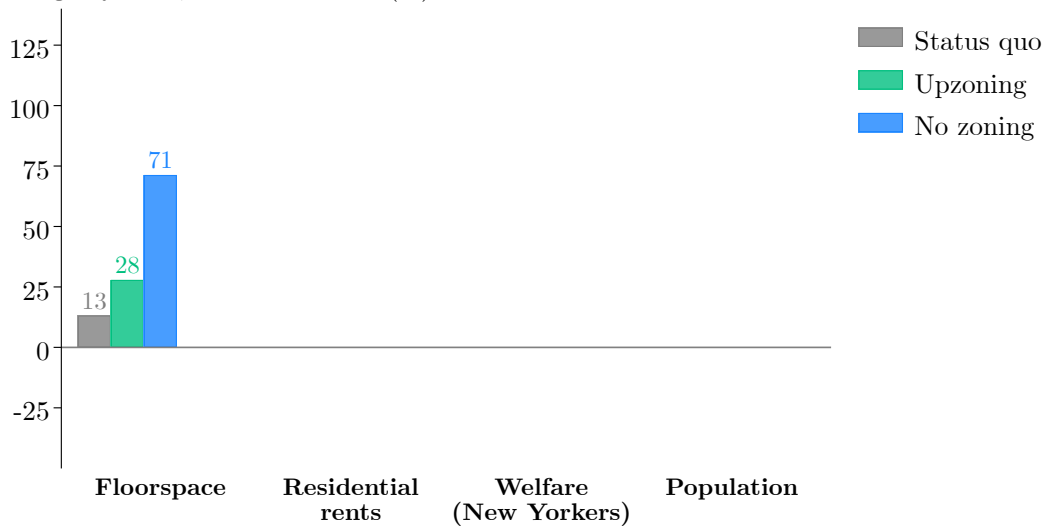
Upzoning increases floorspace by 15pp over 40 years

Change by 2060, relative to 2019 (%)



Completely removing zoning quintuples NYC's growth rate

Change by 2060, relative to 2019 (%)



Rent decreases are moderate

Change by 2060, relative to 2019 (%)



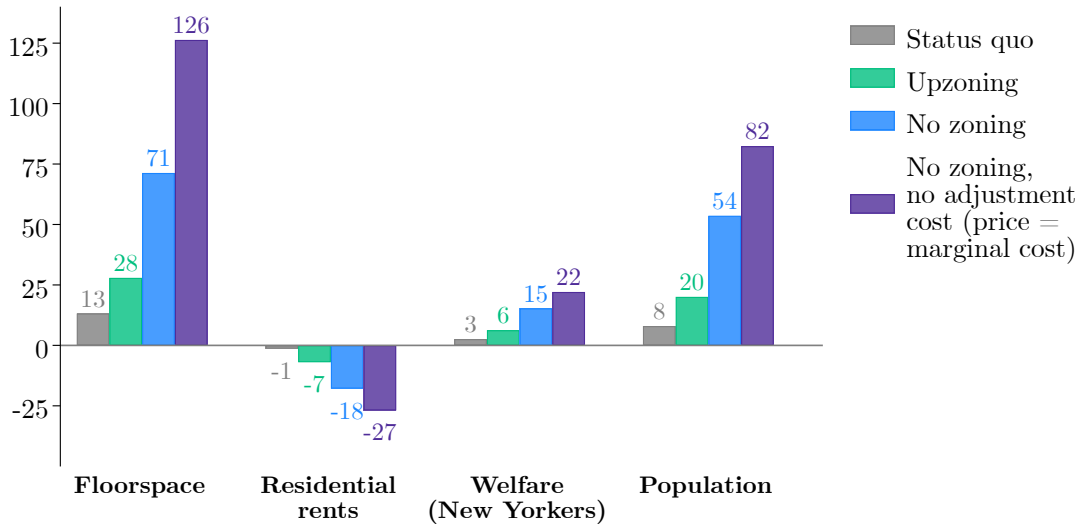
Rent decreases are moderated by migration

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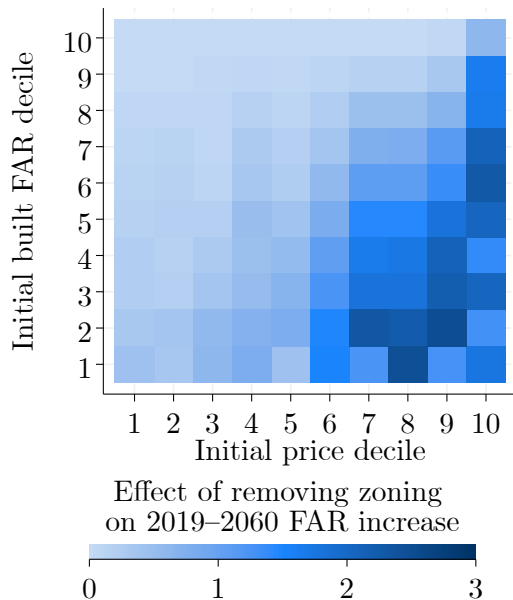


A model ignoring redevelopment greatly overstates effects

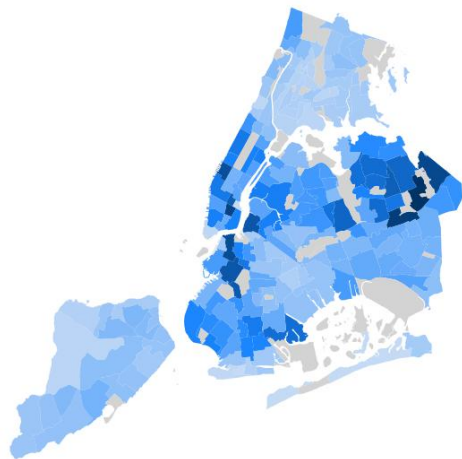
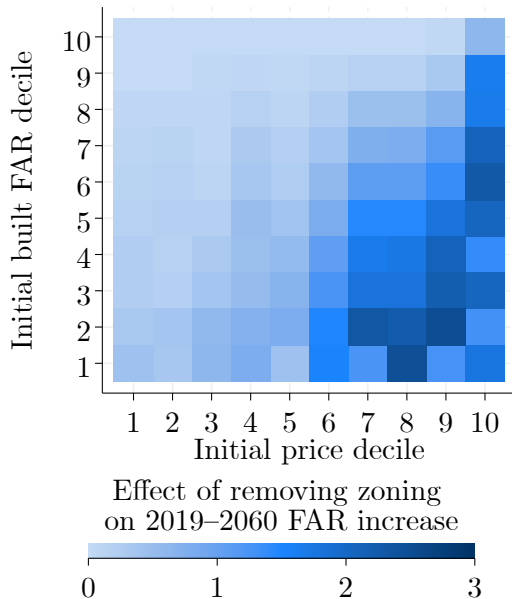
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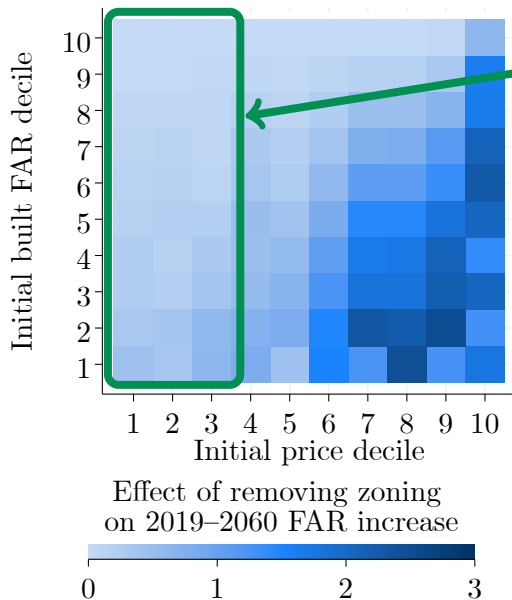
Where does relaxing zoning lead to increased supply?



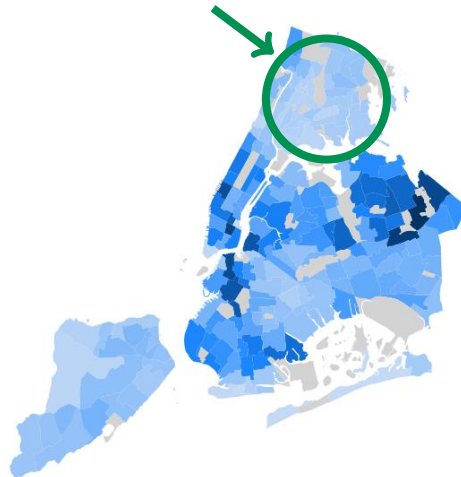
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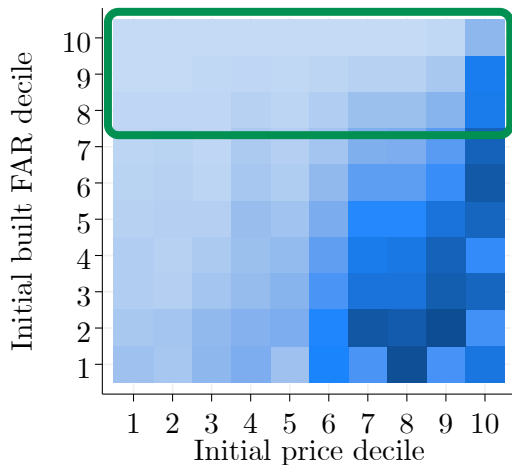
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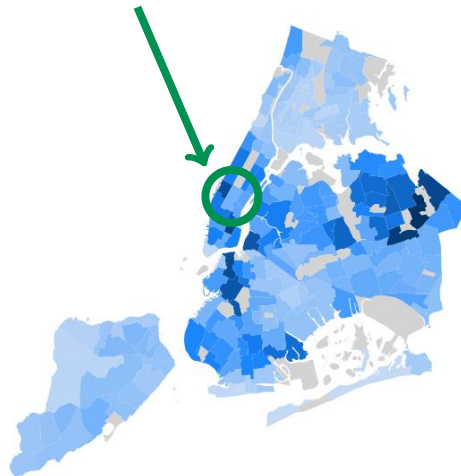
Zoning tends not to be binding when floorspace prices are low.



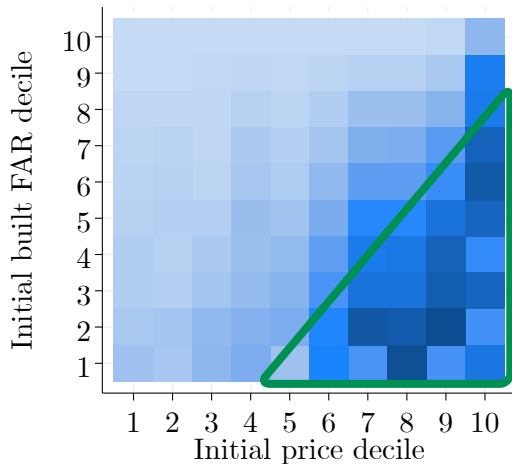
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Removing zoning has little effect on parcels with tall buildings.



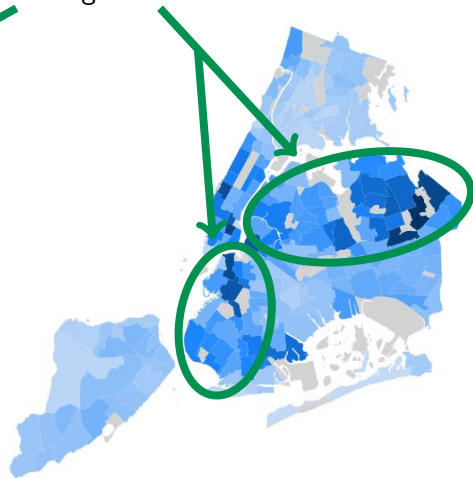
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Effect of removing zoning
on 2019–2060 FAR increase



Upzoning is effective when prices are high enough and density is low enough.



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- **Alternative policies** to promote construction and affordability.
 - Upzoning outperforms reducing construction costs and tax incentives.
- **Supply elasticities.**
 - Compute supply elasticities at a granular level and different time horizons.
 - Supply elasticities vary widely across neighborhoods. Zoning is their primary determinant.

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- Large fixed costs of redevelopment.
 - Even at a 40-year horizon, **take-up of upzoning is limited**.
 - Upzoning works best when **prices are high/density is low**.
- Gains **take time to materialize** and are **geographically spread out**.
 - Local reforms might look disappointing despite **large aggregate/long-term gains**.